

NUMERICAL VERIFICATION OF MTS

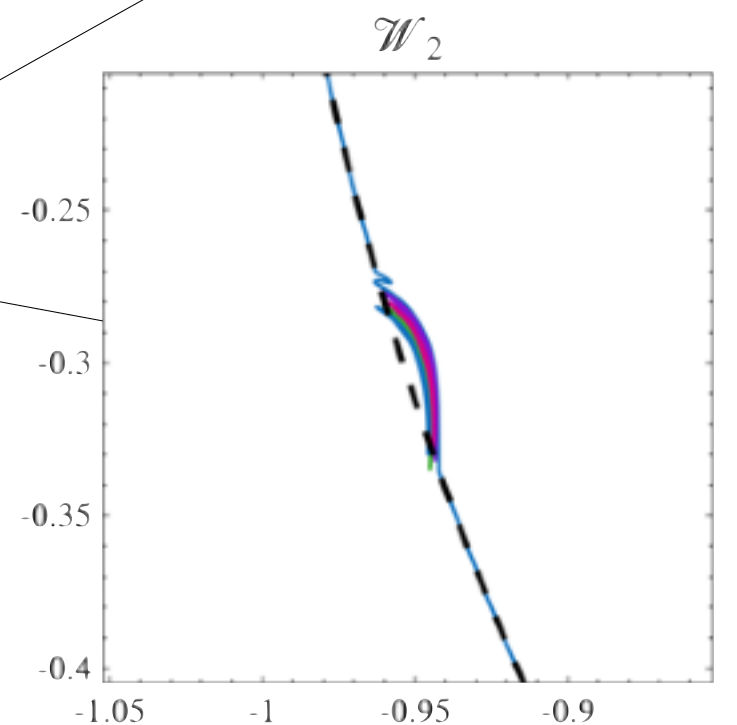
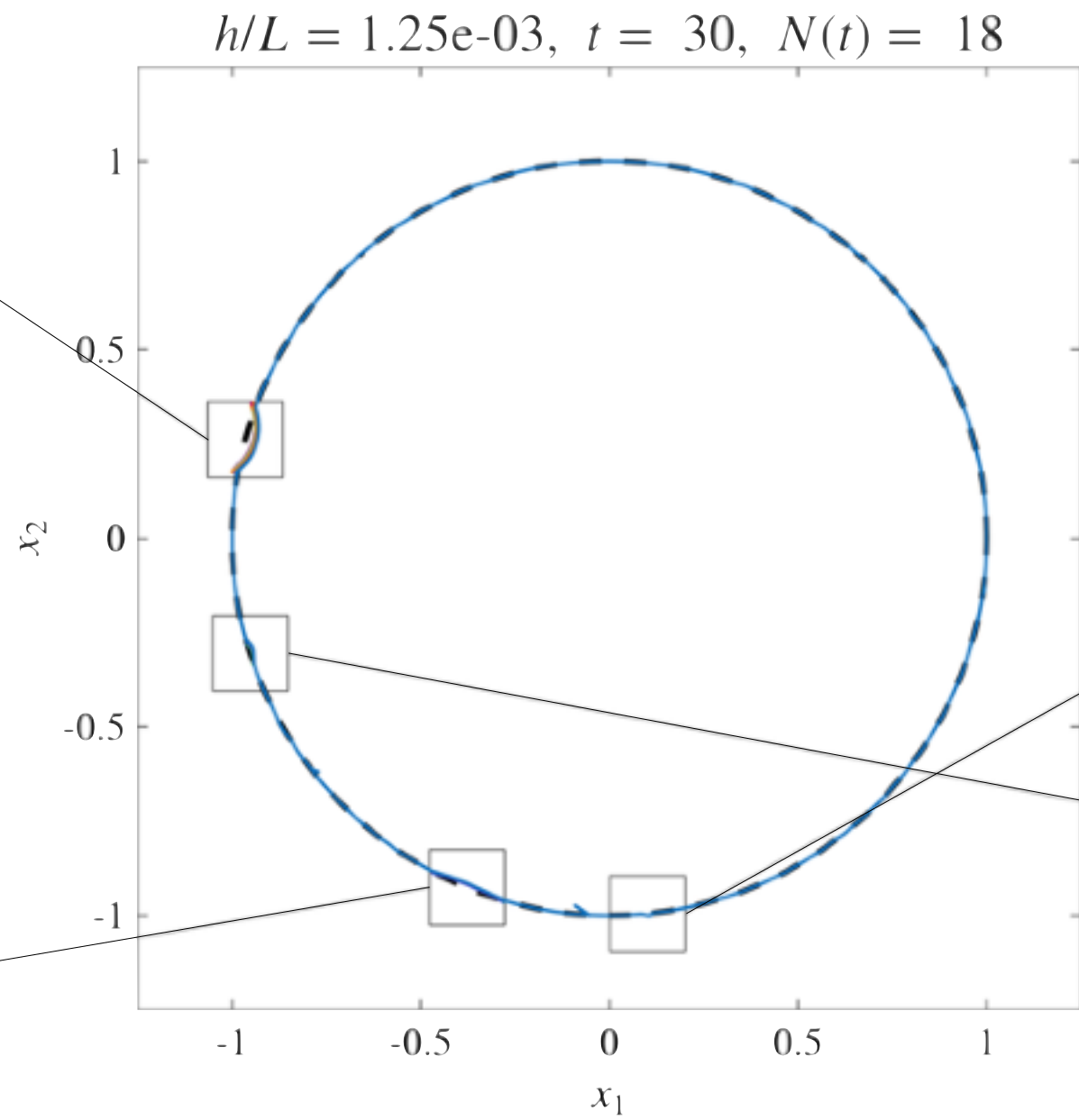
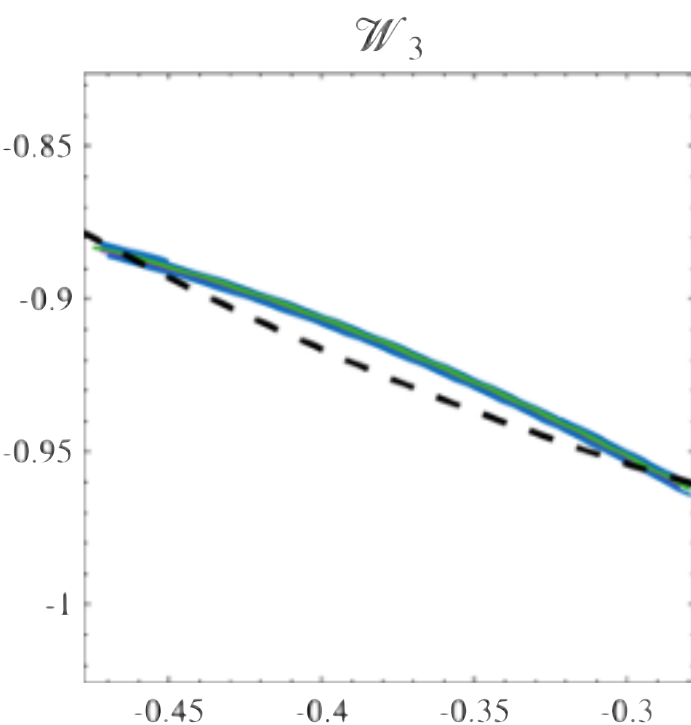
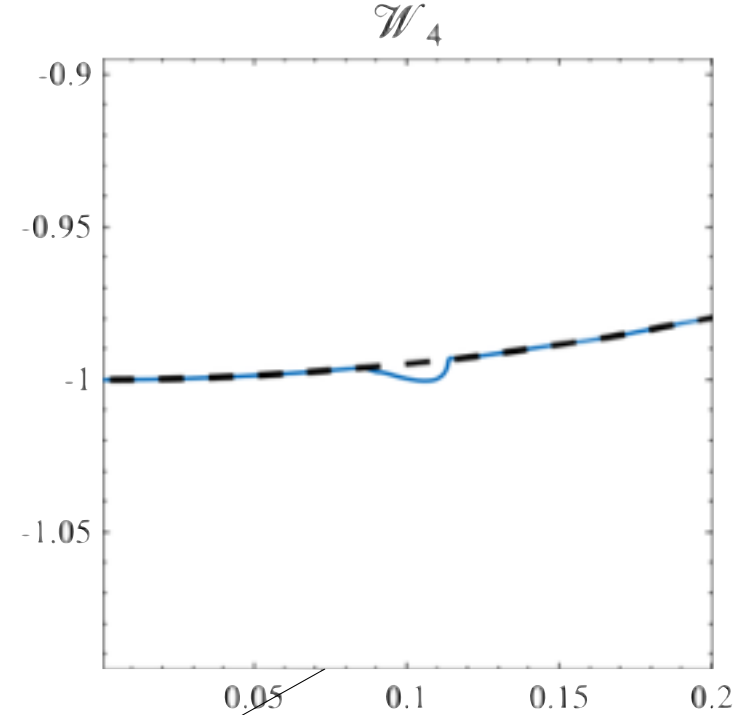
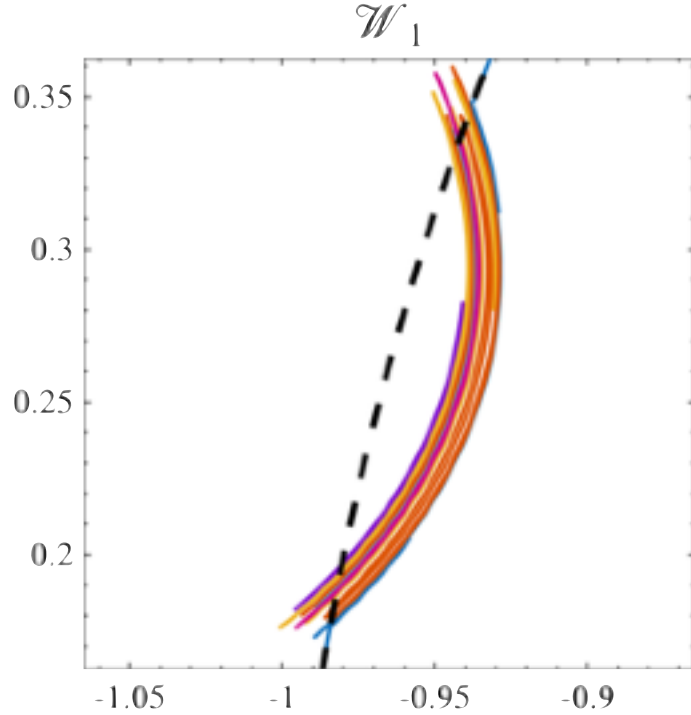


SATELLITE INTERFACES AFTER REVERSAL

After reversal, the remnants of filament breakup localize at the material origins of the topological transitions.

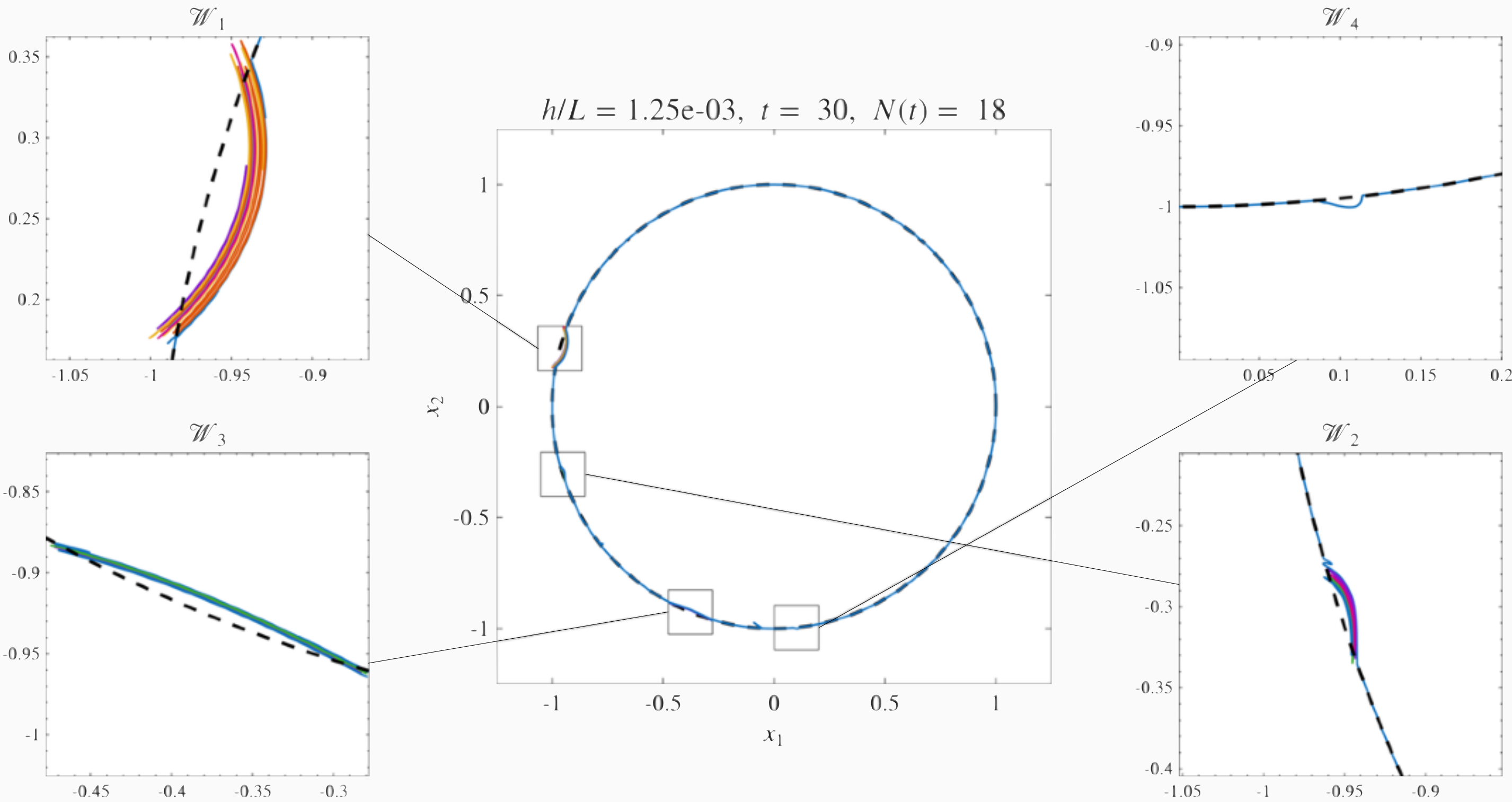
h/L	5×10^{-3}	2.5×10^{-3}	1.25×10^{-3}	6.25×10^{-4}	3.125×10^{-4}
$ A_K - A_0 $	6.92×10^{-3}	1.71×10^{-2}	4.32×10^{-3}	4.76×10^{-4}	7.52×10^{-5}
Order	–	-1.30	1.98	3.18	2.66
E_{sym}	1.63×10^{-1}	4.77×10^{-2}	5.34×10^{-3}	5.92×10^{-4}	7.92×10^{-5}
Order	–	1.77	3.16	3.17	2.90
LGR runtime (s)	4.01	2.04×10^1	7.34×10^1	2.91×10^2	1.11×10^3
Order	–	2.35	1.85	1.99	1.93
MTS runtime (s)	1.58	6.60	2.51×10^1	1.00×10^2	3.73×10^2
Order	–	2.06	1.93	2.00	1.90
Total runtime (s)	5.59	2.70×10^1	9.85×10^1	3.92×10^2	1.48×10^3
Order	–	2.27	1.87	1.99	1.92

CONVERGENCE STUDY. MTS solutions exhibit $\mathcal{O}(h^2)$ accuracy with optimal $\mathcal{O}(h^{-2})$ runtime scaling.



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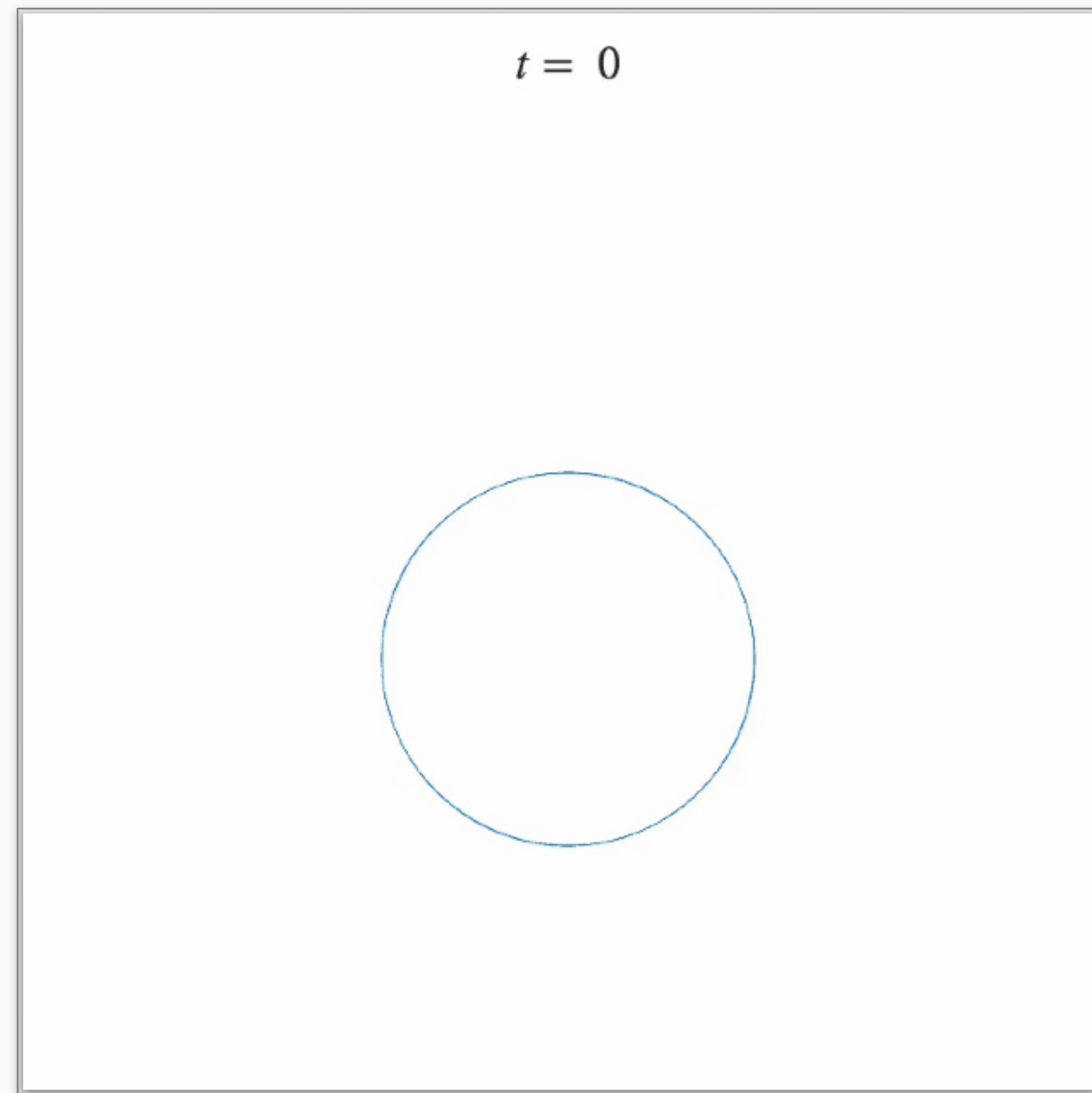
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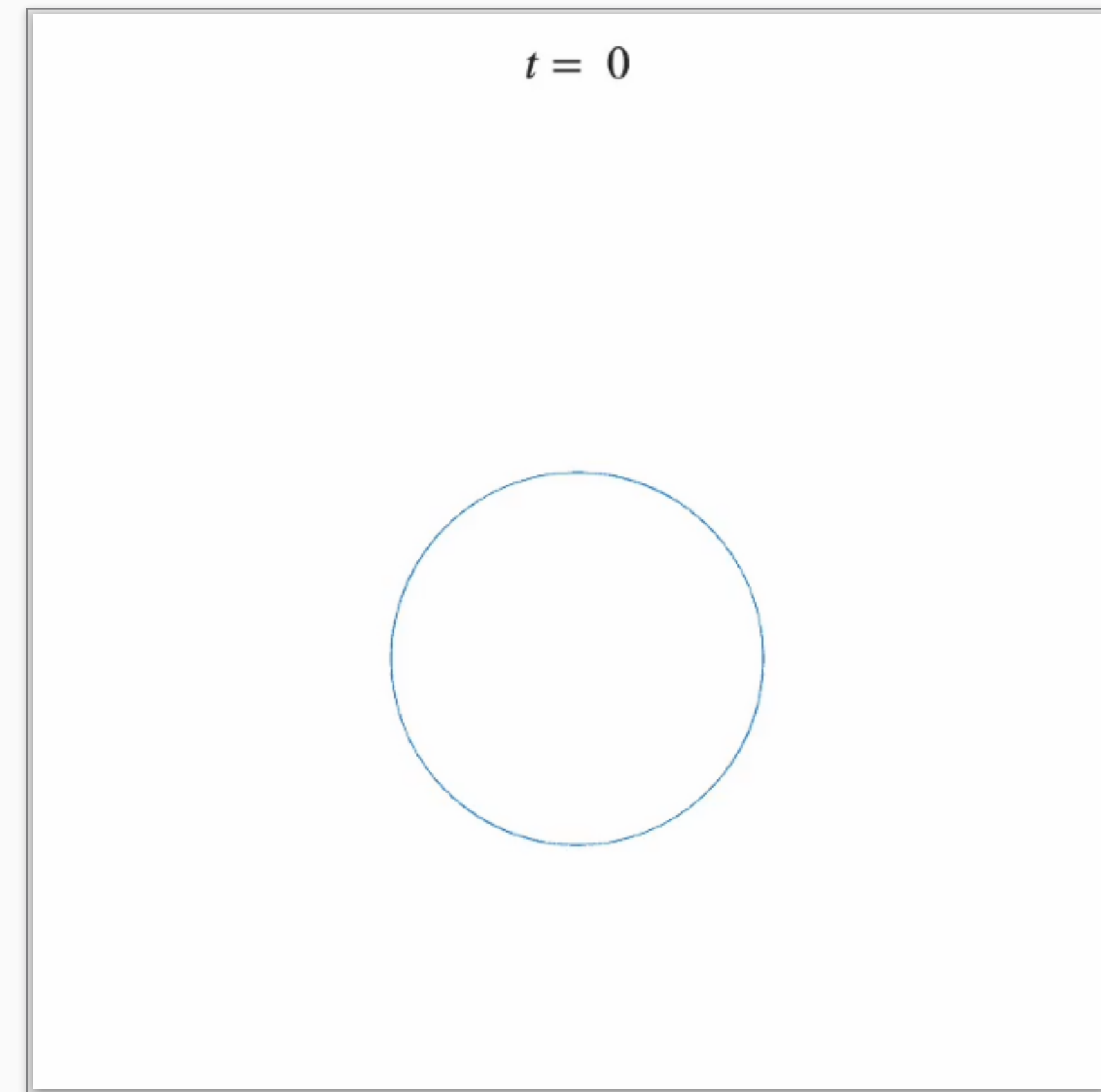
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ENSEMBLE-AVERAGED INTERFACE EVOLUTION



LGR



MTS

ENSEMBLE-AVERAGED INTERFACES. *The left panel shows the classical Lagrangian ensemble, while the right panel shows the corresponding MTS ensemble. Although individual realizations differ substantially, ensemble averaging reveals coherent large-scale interface structures..*